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Final Project Proposal Guide

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Final Project Proposal Guide

1. Introduction

This guide will tell you how to prepare and submit the final project proposal that is the documented work for the Project. A good project proposal must define the functional and non-functional requirements in unambiguous statements, Scope of the Project, Development Schedule, Development Process, Techniques, Tools, Platform with reasoning. However, a professional and well-defined proposal should be composed under the following headings;

a. Project Title
b. Project Overview Statement
c. Project Goals
d. Project Objectives
e. High Level System Components
    a. Component no.1
    b. Component no.2
    c. Component no.3
    d. Component no.4
        i. ……..
f. List of optional functional units
g. Exclusions
h. Application Architecture
i. Gantt chart

1.1 Project Title

The title should be clear and unambiguous (do not make it "cute"). Think of your title as a mini-abstract. A good title should paint a quick picture for the reader of the key idea(s) of your project. The words you use in your title should clearly reflect the focus of your proposal. The most important words should come first, then the less important words. Try to remove words from your title that really are not necessary for understanding. Try and use only a single sentence for your title. If the sentence is getting too long, try removing some words. When all else fails try using a two-part title with the parts separated by a colon (use only as a last resort!). Do not attempt to use the title as an abstract of your entire proposal. If your proposal is built on collaborating with other groups/organizations, it is usually a good idea to include their names on the Title/Cover Page.

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1.2 Project Overview Statement

Think of the Project Overview as an Executive Summary (the busy executive probably only has enough time to read your Overview - not the entire proposal). Be specific and concise. Do not go into detail on aspects of your proposal that are further clarified at a later point in your proposal. The Project Overview should "paint a picture" of your proposal in the mind of the reader. It should establish the framework so that the rest of the proposal has a frame of reference. Use the Project Overview to begin to show your knowledge of the organization from which you are requesting funds. Key concerns of the funding organization can be briefly identified in relation to your proposed project. If you will be collaborating with other organizations, make sure some of their interests are also highlighted in the Project Overview. This can assist in strengthening the collaboration by recognizing them at the very beginning of your proposal. The best time to prepare the Project Overview is after you have completed the entire proposal (and you understand all aspects of your proposal very well). Let the Overview be your last piece of writing and then insert it at the beginning of your proposal. Try to keep in mind that someone will be reviewing your proposal and you would like to have this person be very positive about what you have written. The Project Overview will probably form a strong impression in the mind of the reviewer. Work on your Project Overview so that you can avoid giving this person the opportunity to say things like:

1. Not an original idea
2. Rationale is weak
3. Writing is vague
4. Uncertain outcomes
5. Does not have relevant experience
6. Problem is not important
7. Proposal is unfocused
8. Project is too large.

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Project Overview Statement Template

Project Title:

Group Leader:

Project Members:

Name	Registration #	Email Address	Signature

Project Goal:

Objectives:

Sr.#	
1	
2	
3	
4	
5	
6	

Project Success criteria:

Assumptions, Risks and Obstacles:

Organization Address (if any):

Type of project: ❑ Research ❑ Development

Target End users:

Development Technology: ❑ Object Oriented ❑ Structured

Platform: ❑ Web based ❑ Distributed
❑ Desktop based ❑ Setup Configurations
❑ Other_____

Suggested Project Supervisor:

Approved By:

Date:

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1.4 Project Goals & Objectives

Try and differentiate between your goals and your objectives - and include both. Goals are the large statements of what you hope to accomplish but usually aren't very measurable. They create the setting for what you are proposing. Objectives are operational, describe specific things you will be accomplishing in your project, and are very measurable. Your objectives will form the basis for the activities of your project and will also serve as the basis for the evaluation of your project. Try to insure that there is considerable overlap between the goals and objectives for your proposal and the goals and objectives of the funding organization. If there is not a strong overlap of goals and objectives, then it might be best to identify a different funding organization. Measurable objectives for your project should be presented. If you are dealing with "things" it is easier for them to be measured than if you are dealing with abstract ideas. Your proposal is easier for a prospective funding organization to understand (and the outcomes are much more clear) if you describe your objectives in measurable ways.

The following are the objectives of project:

a- Prepare List of all objectives.

b- --------

c- --------

1.5 High-level system components

Information about the main functional units of the entire system should be present. Functional units to be included will be the inclusive components of the project developed so that the system must perform without taking any physical constraint into consideration. High-level system components are generally, a set of cooperating components assembled together to deliver a solution to a problem. They are frequently identified in terms of inputs, outputs, processes, and stored data that are needed to satisfy the system improvement objectives. If these components are missing the system fails to fulfill its primary mission.

1.6 List of optional functional units

A list of functional units should be present which would include a description of other features, characteristics, and constraints that define a satisfactory system. These functional units would be developed under certain conditions (technology, expertise, or time dependent). Examples of these optional functional units would include performance (throughput and response time); ease of learning and use; budgets, costs, and cost savings; timetables and deadline; documentation and training needs; quality management; and security and internal auditing controls.

They are often requirements that specify need of compliance with any legal and regulatory requirements. They can also be design constraints due to the operating system used, the platform environment, compatibility issues, or any application standards that apply. In general, you can say that any requirement that does not allow for more than one design option should be regarded as a design constraint.

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If the optional functional units are missing the system can still (for a while) fulfill its fundamental mission, but with degraded service quality.
While gathering and validating the optional functional requirements, maintain Assumptions and Issues lists.

Some activities will not give you satisfactory answers. This can be due to lack of information, or simply because you consider the answer threatens the viability of the design. Therefore, create two lists, and maintain them through the design study:

Any assumptions you make during the requirements and design process, including the rationale or thought processes behind those assumptions. Assumptions may be used to identify related subprojects or items of work, which are outside the scope of or after this project any major issues (significant concerns that could become show-stoppers).
The issues should be reviewed with the customer at the end of each phase. The assumptions need to be reviewed also, at the end of each phase, but the customer might not always be the correct person for the less important ones. Assumptions and issues apply to all artifacts, but are particularly common for non-functional requirement.

1.7 Exclusions

A list of the functional units, which will not be intended to be develop or discussed during any point in the project development, should be present. Time constraints or lack of resources for the fulfillment of the required task or any sort of other constraint preventing the completion of the functional unit could be described here.

1.8 Application Architecture

Defines the overall application architecture e.g. a two-tier architecture or a three-tier architecture. It must contain a diagram depicting the system architecture properly
Architecture is the highest-level concept of a system in its environment. The architecture of a software system (at a given point in time) is its organization or structure of significant components interacting through interfaces, those components being composed of successively smaller components and interfaces.
Architecture can also be defined as the organizational structure of a system. Architecture can be recursively decomposed into parts that interact through interfaces, relationships that connect parts, and constraints for assembling parts. Parts that interact through interfaces include classes, components and subsystems.
There are a number of typical patterns of distribution in systems, depending on the functionality of the system and the type of application. In many cases, the distribution pattern is informally used to describe the 'architecture' of the system, though the full architecture encompasses this but also many more things. For example, many times a system will be described as having’ client-server architecture', although this is only the distribution aspect of the architecture.

1.9 Gantt chart

The Gantt chart enumerates the activities to be performed on the vertical axis and their corresponding duration on the horizontal axis. It is possible to schedule activities by

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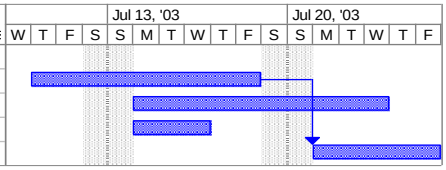
either early start or late start logic. In the early start approach; each activity is initiated as early as possible without violating the precedence relations. In the late start approach; each activity is delayed as much as possible as long as the earliest finish time of the project is not compromised.

Based on the Work Breakdown Structure (WBS), a timeline or Gantt chart showing the allocation of time to the project phases or iterations should be developed. This Gantt chart would identify major milestones with their achievement criteria. It must contain duration estimation of all the necessary activities to be carried out during the project development along with the human resources responsible for the respective tasks. Activity dependencies are also required to be mentioned in it.

Sample Gantt chart

ID		Task Name	Duration	Start	Finish	Predecessors
1						
2		billing	7 days	Thu 7/10/03	Fri 7/18/03	
3		computing	8 days	Mon 7/14/03	Wed 7/23/03	
4		accounting	3 days	Mon 7/14/03	Wed 7/16/03	
5		marketing	10 days	Mon 7/21/03	Fri 8/1/03	2



1.10 Hardware and Software Specification

Any hardware or software specifications e.g. machine type required, operating system and other utilities should be clearly specified for the system to be developed.

1.11 Tools and technologies used with reasoning

The application tools, which are to be used on front and back end of the system to be developed, should be listed. The reasons for these tools should also be enlisted.
Identify what the needs for tool support are, and what the constraints are, by looking at the following:

Designing Tools:

- Rational Rose for UML diagrams can be one example.
- The development process. What tool support is required to effectively work? For example, if the organization decide to employ an iterative development process, it is necessary to automate the tests, since you will be testing several times during the project.
- Host (or development) platform(s).
- Target platform(s).
- The programming language(s) to be used.

Development Tools:

- Eclipse for java
- Existing tools. Evaluate any existing and proven tools and decide whether they can continue to be used.

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- The distribution of the development organization. Is the organization physically distributed? Development tools generally support a physically distributed organization differently.
- The size of the development effort. Tools support large organizations more or less well.
- Budget and time constraints

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Final Documentation Format Guidelines

Title Cover Specimen
Hard Bound
(After 3 Enter Spaces)

TYPE TITLE OF FINAL PROJECT HERE

(16 pt. Bold, Capital Letters, Single Space, Align Centre)
(After 5 enter Spaces)

(After 4 enter spaces)

Project ID- 00

Project Advisor: Name of Project Adviser

Submitted By

Name of Student	ID of Student
Name of Student	ID of Student

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Inner Title Cover Specimen
(After 3 enter spaces)

Type Your Project Title Here

(14 pt. Bold, Capitalized Each word, Align Centre)
(After 6 enter Spaces)

BS in Information Technology-20XX
MSc in Information Technology-20XX
(14 pt. Bold, Capitalized Each word, Align Centre)
(After 6 enter times)

A project submitted in partial fulfillment of the
requirements for the award of the degree of
BS in information Technology /
MSc in Information Technology
(After 11 enter times)

NAME OF DIVISION/CAMPUS

UNIVERSITY OF EDUCATION

LAHORE

November 2018

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"I hereby declare that I have read this project documentation and in my opinion this project is sufficient in terms of scope and quality for the award of the degree of BS in Information Technology/MSc in Information Technology."

_____

Project Primary Supervisor
Name: Write name of Project advisor here
Designation: Write designation of Project Advisor here
University of Education.

_____

Project Examiner
Name: Write name of Project Examiner here
Designation: Write designation of Project Examiner here
University of Education.

DECLARATION

I declare that this project title entitled "**write your project title here**" is the result of my own research and development except as cited in the references. This project has not been accepted for any degree and is not concurrently submitted in candidate for any other degree. At any time if my statement is found to be incorrect even afterwards of BS in Information Technology/MSc in Information Technology, the university has the right to withdraw my BS in Information Technology/MSc in Information Technology degree.

Signature: _____

NAME IN CAPITAL LETTERS
Name: _____

Month date, year
Date: _____

Signature: _____

NAME IN CAPITAL LETTERS
Name: _____

Month date, year
Date: _____

Signature: _____

NAME IN CAPITAL LETTERS
Name: _____

Month date, year
Date: _____

Signature: _____

NAME IN CAPITAL LETTERS
Name: _____

Month date, year
Date: _____

PLAGIARISM UNDERTAKEN

I solemnly declare that project work presented in this documentation entitles **"Name of your project**" is solely my work with no significant contribution from any other person. Small contribution/help wherever taken has been acknowledged and that complete project has been written by me.

I understand that zero tolerance policy of the HEC and University of Education, Lahore towards plagiarism. Therefore, we as an author of the above titled project declare that no portion of my project documentation and any material used as reference is properly referred/ cited.

I undertake that of I am found guilty of any formal plagiarism in the above titled project even after award of BS/MSc degree, the University reserve the rights to withdraw/revoke my BS/MSc degree and that HEC and the University has the right to publish my name on the HEC/University Website on which names of students are place who submitted plagiarized projects.

Signature: _____

NAME IN CAPITAL LETTERS

Name: _____

Month date, year

Date: _____

Signature: _____

NAME IN CAPITAL LETTERS

Name: _____

Month date, year

Date: _____

Signature: _____

NAME IN CAPITAL LETTERS

Name: _____

Month date, year

Date: _____

Signature: _____

NAME IN CAPITAL LETTERS

Name: _____

Month date, year

Date: _____

CERTIFICATE OF APPROVAL

This is to certify that the project work presented in this documentation entitled, "**write name of your project**", was conducted by **"name of member 1", "name of member 2", "name of member 3", "name of member 4"**, under the supervision of **"write your supervisor name"**. No part of this project has been submitted anywhere else for any degree. This project is submitted to the **"name of your campus, University of Education"** is partial fulfillment of the requirements of the degree of BS in Information Technology/MSc in Information Technology.

Signature: _____

NAME IN CAPITAL LETTERS
Name: _____

Month date, year
Date: _____

Signature: _____

NAME IN CAPITAL LETTERS
Name: _____

Month date, year
Date: _____

Signature: _____

NAME IN CAPITAL LETTERS
Name: _____

Month date, year
Date: _____

Signature: _____

NAME IN CAPITAL LETTERS
Name: _____

Month date, year
Date: _____

_____
Project Primary Supervisor
Name: Write name of Project advisor here
Designation: Write designation of Project Advisor here
University of Education.

_____
Project Examiner
Name: Write name of Project Examiner here
Designation: Write designation of Project Examiner here
University of Education.

OFFICE OF CONTROLLER OF EXAMINATION

NOTIFICATION

No: _____ Date: _____

It is notified for the nomination of all the concerned that Mr./Ms. (name of the student-1), (name of the student-2), (name of the student-1), (name of the student-1) BS/MSc student of Name of the department of University of Education has completed all the requirements for the award of BS/MSc Degree in the discipline of Information Technology as per detail given hereunder:

BS in Information Technology/
MSC in Information Technology

		Cumulative Result			
		Credit Hours:_____			**Cumulative Grade Point Average (CGPA)**
Registration No	**Complete Name**	**Course work**	**Project**	**Total**	

Project Title: _____

Name of Supervisor: _____

Signed by

Controller of Examination

CC:
1.abcd
2. xyz3

ACKNOWLEDGEMENT

We truly acknowledge the cooperation and help make by Name of **Acknowledger**, **Designation** of **Address of Organization**. He has been a constant source of guidance throughout the course of this project. We would also like to thank **Acknowledger** from **Designation**, **Address of Organization** for his help and guidance throughout this project. We are also thankful to our friends and families whose silent support led us to complete our project.

1- Mr. Furqan

2- Mr. Akram

Date:

March 11, 2016

ABSTRACT

A good abstract explains in a few lines what was actually done and why the work was important. The maximum number of words should not be more than 1000 words. The abstract page may include the following:

- Background
- Objectives/aim of the study
- Research methodologies
- Findings
- Conclusions
- Implications
- Limitations

DELIVERABLE SUBMISSION GUIDELINE

Name of Deliverable	Contents	Remarks
Deliverable-I	**Project Proposal** **Chapter-1:** Gathering & Analyzing Information **Chapter-2:** Software Requirement Specification (Sample Attached)	Deliverable-I, should be submit after the commencement of 7th Semester.
Deliverable-II	**Chapter-3:** Analysis **Chapter-4:** Design **Chapter-5:** Graphical User Interfaces **Chapter-6:** Testing **Chapter-7:** Conclusion and Future work **Complete Running Application**	Deliverable-II, should be submit after the commencement of 8th Semester.

TABLE OF CONTENTS

(Required Chapters of Project Documentation)

TITLE AND DESCRIPTION:

- Inner Title Page
- Statement of Submission
- Declaration
- Plagiarism undertaken
- Certificate of Approval
- Notification
- Acknowledgement
- Abstract

CHAPTER NO. 1: Gathering & Analyzing Information

- Introduction
- Problem Statement
- Goal & Objectives
- Research Questions
- Methodology
  - Available Methodologies
  - Chosen Methodology
  - Reasons for Chosen Methodology
- Definitions, Acronyms and Abbreviations

CHAPTER NO. 2: Software Requirement Specification (Sample Attached)

- Stakeholders Characteristics
- Domain Requirements
- Functional Requirements
- Non-Functional Requirements

CHAPTER 3: Analysis [Use Case Description and Use Case Model] (Sample Attached)

CHAPTER 4: Design [with Description of each diagram]

- Architecture Diagram
- ERD
- Data Flow diagram (Level 0 and 1)
- Class Diagram
- Sequence Diagram

CHAPTER 5: Graphical User Interfaces

- (Mockups of working software application)

CHAPTER 6: Testing

- Introduction
- Test Scenario **(Sample Attached)**
- Test Plan
- Definition of Test Cases
- Test Cases Specifications
- Test Cases Results for:
    - Black Box Test Cases
    - White Box Test Cases.

CHAPTER 7: Conclusion and Future work

References (APA 6 Edition Style)
Appendix

Appendix: Final Documentation Format Guidelines

Typographical Format and Binding

Color of Project Documentation Binding

Name of the Degree Program	Text Color	Color of Binding
BS in Information Technology	**Black with Silver Script**	ABC
MSc in Information Technology	**Navy Blue with Gold Script**	ABC

Page Format:

Page size:	A4
Top margin:	1.00 inch
Bottom margin:	1.00 inch
Left margin:	1.5 inch
Right margin:	1.00 inch
Page numbering:	Bottom right - part of the footnote Title page not numbered All other pages before the page of chapter one numbered in lower roman numerals (i, ii, iii, …) All other pages starting from first page of chapter one to last page of the report numbered in integers (1, 2, 3, …)
Footer:	Each page shall have a footnote "University of Education" Left aligned In case of long titles shorter versions should be used. There shall be a line over the footnote.
Header:	Each page shall have a header "Project Name" Left aligned In case of long titles shorter versions should be used. There shall be a line under the footnote.
Chapter Startup:	Each chapter shall be numbered as Chapter 1, Chapter 2, etc. The name of the chapter shall be written immediately below. Both shall be centered horizontally as well as vertically. The actual chapter content shall start from the next page.

Text:	Only one side of the paper shall be used. The other side shall be blank. When a report is opened the right side would contain text, figures, or tables and the left side would be blank.
Tables and Figures:	Tables and figures shall be placed on one side only Separate pages shall be used for figures and tables. One page may contain more than one figure or table but text will not be combined or interlaced with figure or table. Each table / figure shall be numbered. For example, "Table 1.2: Population distribution in Asia" or "Figure 3.2: Temperature distribution" The table number or figure number shall be placed as normal text centered at the bottom of the table or figure or sideways with table / figure title coming on the opening side of the paper and note on the binding side.

Paragraph:

Single-spaced.
Line entered paragraph.
DONOT put indents at the beginning of the paragraph.
Left aligned or justified.

Text Format

Normal and plane text:

Font Type:	Times New Roman
Font Size:	12

Headings:

Chapter Heading:	Times New Roman Bold Size 16 Title Case normal
Heading 1:	Times New Roman Bold Size 14 Title Case normal
Heading 2:	Times New Roman Bold Size 12 Title Case normal
Heading 3:	Times New Roman Bold Size 12 Title Case italic

Sections and Subsections

In case of sections and subsections follow this format:

1 Section
1.1 Sub Section
1.1.1 Nested Sub Section
    a
    b
        i
        ii

The subsequent reference to a any section shall be made using the section and its number. For example, **section 2.1.3** means chapter 2 section 1 subsection 3.

Mathematical Equations

The following numbering scheme should be used to number the equations:

$f(x) = x+3$ (XX:YY)

Where XX is the chapter number and YY is the sequence number of that equation in that chapter.

If an equation is previously quoted in an earlier chapter, say as equation 4:5 and need to be re-quoted in chapter 5, its number will remain as equation 4:5.

References

References are to be placed in square brackets and interlaced in the text. For example, "A comprehensive detail of how to prevent accidents and losses caused by technology can be found in the literature [1]. A project report / thesis cannot be accepted without proper references. The references shall be quoted in the following format:

The articles from journals, books, and magazines are written as:

[1] Abe, M., S. Nakamura, K. Shikano, and H. Kuwabara. Voice conversion through vector quantization. *Journal of the Acoustical Society of Japan*, April 1990, E-11 pp 71-76.

[2] Hermansky, H. Perceptual linear predictive (PLP) analysis for speech. *Journal of the Acoustical Society of America*, January 1990, pp 1738-1752.

The books are written as:

[1] Nancy G. Leveson, Safeware System Safety and Computers, A guide to preventing accidents and losses caused by technology, Addison-Wesley Publishing Company, Inc. America, 1995.

[2] Richard R. Brooks, S. S. Iyengar, *Multi-Sensor Fusion Fundamentals and Applications with Software*, The Prentice-Hall Inc. London, 1998.

The Internet links shall be complete URLs to the final article.

[1] http://www.ue.edu.pk

For Softcopy Submission

Contents:

All reports / theses must accompany a CD whose contents will have the following:

Top-level directories:

Doc	All documents related to the project Instructions how to access the software to the point to running the project

All reports already submitted
The final project report in thesis form
Installation instructions
Trouble shooting instructions in case of problems
User manual
Research material including URLs
Papers consulted / referred to
Slides of the presentations

Source — All source files that will be needed to compile the project.
Further subdirectories can be used.
This must include sample data files as well.

Project — The running project including sample data files as well as sample output.
This should be in a form that if copied to a machine runs without errors.
This may an exe file of an entire project, an installer depending on the project or simply a running project.

You can have sub directories with appropriate names.

Length

The length of your dissertation depends on the type of project you have selected. An excellent dissertation will often be brief but effective (its author will have said a lot in a small amount of space). Voluminous data can be submitted electronically on CD.

Sample for Software Requirement Specification

.

Academic Automation System
(Specimen case study)

Software Requirements Specifications

Functional Requirements

No	Requirement	Description
FR1	Create Semester	The Academic Manager can create a new semester for offering. The semester is automatically offered to students when the Semester Registration Start Date is reached. The following information will be needed: • Year • Type (Spring, Summer, Fall) • Semester Start Date • Semester End Date • Semester Registration Start Date • Semester Registration End Date • Course Registration Start Date • Add Course Last Date • Drop Course Last Date • Withdraw Course Last Date • Classes Start Date • Classes End Date
FR2	Offer Course	The Academic Manager can offer a previously created course in the semester. The information provided will be: • Course (Title + Code) • CGPA Requirement • Registration Type (Primary Sections/Open) Depends on FR1.
FR3	Create Section	**Sections have to be created when a course is offered to students. The Academic Course Coordinator enters the following information must be provided:** • Instructor • Registration Last Date • Add Last Date • Drop Last Date

		• Withdraw Last Date • Class Timings • Registration Type (Batch-wise/Open) • Seats Distribution (if Batch-wise registration) Depends on FR1 and FR2.
FR4	Semester Registration Request	The student can register in the current semester. This includes a transaction equal to the Registration Fees in his account. A student can register in one semester at a time.
FR5	Add Student to Course	When a student's requests to add a course, the system will check the number courses that the student is already registered in and student's GPA against the requirement of the course. For BS and MS, maximum limit of courses in a semester is 5. The system will then check for clashes between the registered courses and the new course. If a student wants to take 6 courses, the advisor will have to approve this. A student with CGPA below 2 also needs approval from the advisor. The advisor 's approval should be received within 2 days of the registration, if registration does not close within those days. Both BS and MS students must have successfully cleared all pre-requisites of the courses that he/she is taking. The student must have cleared all previous dues to register for the new semester. The system will check the number of seats available. BS students can take MS courses and vice versa, if they are offered to them. However, BS students require the course instructor's approval for taking a MS course. MS students require the approval of their advisor/coordinator for taking any course. If all checks are validated the students is registered in the course.

		If the advisor's approval is required, then the student has 2 days for getting the approval (if the last date of registration is not in those 2 days) and a further 2 days (if the last date of registration is not in those 2 days) to submit his fees and get his name finally registered, else his name will be automatically removed. If seats are filled he/she is added to the waiting queue. The student can be added in 3 courses for the waiting list at a time.
FR6	Drop Course	The student can request to drop from any course he is registered in until the add/drop date. If there are students in the waiting list for the course, the next student is notified automatically. The students of the batch to who this course was offered are given priority. The new student has 2 days to confirm this registration by paying his dues, else his name is dropped from the waiting list, the list is updated and the next person is notified. Students with CGPA below 2 cannot drop a course without approval from the advisor. Depends on FR7.
FR7	Withdraw Course	The student can request to withdraw from any course he is registered in until the withdraw date. The status is changed to 'W' (withdraw) to appear on the transcript. Students with CGPA below 2 cannot withdraw from a course without approval from the advisor. MS students cannot withdraw from a course. Depends on FR7.
FR8	View reports	The various types of reports that the users (depending on their access level and role) can view are: • Course-wise Registration (specifying sections)

		• Student-wise Registration (specifying sections) • Waiting List of that Course (specifying sections) • Date-wise Add/Drop status of each student – This report will be delivered after the rest of the system, by 15th January 2002.
FR9	Semester Freeze	A student can freeze his next or previous semester by paying the required amount to Accounts. The fee has to be paid for each semester (maximum 2) that is to be frozen. The current semester cannot be frozen if a student is taking courses.
FR10	Clear Accounts Status	The Accounts Officer can clear the status of a student when he pays his dues. When the student pays his dues or submits the financial aid/installments application, the Accounts Officer clears his financial status so that the student's seat can be confirmed. When the student pays his semester freeze dues the Accounts Officer does the same.

Non-Functional Requirements

No	Requirement	Description
NFR1	Security	• All access to the system will be logged. • Only an administrator shall be able to create users. • Only authorized and authenticated users shall be able to access the system i.e. a user with valid username and password and the right post/domain. • System operators shall not have access to the data that they do not need • Database shall be accessed indirectly using predefined queries so that there is no direct SQL injection attack i.e. the user information that may affect the queries generated for the system or corrupt database • Illegal operations may lead to termination of program. Error handling and exceptions will stop any illegal operations.

NFR2	User Interface	Internet explorer 6.0+ will work as the client application on Windows.
NFR3	Platform Independence	The system will be independent of a single platform and commercial software.
NFR4	Extensibility	The system will be extensible and modular to integrate with other automated systems as they are developed
NFR5	Performance	The system will be able to handle multiple requests over the Intranet of 50 users at the maximum. When the applicant submits his information, the maximum delay before getting a response shall not exceed 3 minutes The availability of the application form shall be 24 hours a day during the dates it is uploaded on the site. The response time experienced by the administrator while editing or uploading the application form shall not be more than 30 seconds. Accuracy of up to 4 decimal places shall be provided while generating reports The software shall produce reports on the data saved in the database. Therefore, the software has the capacity to generate reports of any size that is within the database limits. After a training of 1 hour of an experienced computer user, the average number of mistakes made by any system operator shall not exceed 1 per day
NFR6	User Documentation	Online help shall be made available for: a) The applicants to help them in following the application procedure. b) The candidates giving the test to guide them through the instructions at the start of the test. • Reference manual shall be provided for all system operators defining user interfaces and functionalities provided by each interface to assist them in making use of the functionalities provided. • Installation manual shall be provided to the system users. • Recovery manual shall also be provided to assist them in case of system failures.

Use Case & Test Case Sample

.

USE CASE SCENARIO SAMPLE

UC Number: 1.1

UC Name: Login

Functional Requirement No: FR1

Primary Actors/Stakeholders: Student, Teacher

Secondary Actors/Stakeholders: Admin

Description: To get admin authentication and logged, the users will be prompted to login with their account information before they can use the system.

Preconditions: The user must have valid account.

Main Success Scenario (MSS):

1. The user connects to the system.
2. The user enters his/her username and password.
3. The system validates the username and password.
4. The system determines the user's role.
5. The system displays a list of actions the user can perform based on the user's role.

Alternative Scenario:

1. Invalid account user or pass
2. User already logged into the system

Post conditions:

1. The admin will log in to the system
2. The admin has access to the functions of the system

Extensions:

1. The system determines that the password is incorrect for the username entered.
    a. The system prompts the user to re-enter the password.
        i. The system determines that the re-entered password is incorrect.
        ii. The system provides the option for the user to retrieve a forgotten password.

2. The system determines that the username does not match a username for any account.
    a. the system displays an error message.
3. The system determines that the user has no role assigned in the system.
    a. The system does not allow the user to access the system.

TEST CASE SCENARIO SAMPLE

Test Case ID : 01

Test Case Name: Login

Test Priority: Medium/ High/ Low

Preconditions: The user must have valid account.

Post conditions:

1. The user will log in to the system
2. The user will have access to the functions of the system

SN	Action	Inputs	Expected Outcome	Actual Output	Test Application	Test result	Test comments
1	Launch Application	http://lms.ue.edu.pk	LMS home page	LMS home page	Internet Explorer-15	PASS	[Ali 10/0/2018]: Launch Successful
2	Enter correct email and password	Email ID: test@ue.edu.pk Password: ****	Login Success	Login Success	Internet Explorer-15	PASS	[Ayesha 10/0/2018]: Launch Successful

RULES AND REGULATIONS
FOR
FINAL YEAR PROJECT (FYP) EVALUATION (VERSION-I)
DEPARTMENT OF INFORMATION SCIENCES
UNIVERSITY OF EDUCATION, LAHORE,

In pursuance of meeting and direction by the Chairperson Department of Information Sciences, the following Rules and Regulations are prescribed to administer the affairs concerning to the Final Year Project(FYP) Evaluation, Department of Information Sciences.

Short Title and Commencement

This document shall call Rules and Regulations for Final Year Project (FYP) Evaluation (Version-I), Department of Information Sciences, University of Education, Lahore, Pakistan.

a) These Rules and Regulations shall apply to the students admitted in BS and MSc degree program offered by the Department of Information Sciences, University of Education right after approval from the BOS and BOF to all existing and future batches unless modified.

1) Definitions

In these Rules and Regulations, unless the context otherwise requires, the following expressions shall have the meanings hereby respectively assigned tothem;

"University" means the University of Education, Lahore,Pakistan.

a) "UE" stands for University of Education.

b) "Vice Chancellor" means the Vice Chancellor of theUniversity.

c) "Department' means an Academic Department for teaching, research, and technological development of Law maintained and administered by theUniversity.

d) "BoS" stands for Board of Studies

e) Authority" means any of the Authorities of the University specified in section 18 chapter IVof University of Education Ordinance, 2002.

f) "FYP" stands for Final YearProject

g) "CE" stands for Controller of Examinations, University ofEducation.

h) "CH" stands for Credit Hours.

i) "DAI" stands for Degree Awarding Institute recognized by HEC,Pakistan.

j) "DFYPC" stands for Departmental Final Year Project Committee notified by the Director/Principal concerned as prescribed in these Rules andRegulations.

k) "PMO" stands for Project Monitoring Officer to be nominated by Chairperson/HOD/Coordinator concerned and notified by Director/Principal Concerned.

l) "BS" stands for Bachelor Studies

m) "Chairperson/HOD/Coordinator" means Chairperson/Head of Department/Coordinator of the Academic Department.

n) "College" means a constituent college or an affiliatedcollege.

o) "Dean" means the Chairperson of the Board ofFaculty.

p) "Final Year" means 08th Semester of the BS degreeprogram or 4th Semester of MSc degree program.

q) "Project" means an individual or collaborative enterprise that is carefully planned to achieve a particularaim.

r) "Project External Examiners" means examiners, appointed by PMO to evaluate the Thesis/Project and conduct the Viva Voce.

- The Project External examiners shall be within campus or inter UE campuses.

s) "Faculty" means a Faculty ofUniversity.

t) "Supervisor" means a Full Time Regular / IPFP / TTS / Long Terms Contractual Faculty member of the department, University of Education having enough time to mentor the student project till completion of the Project. He/She shall be appointed by the Departmental Final Year Project Committee (DFYPC).

u) "Faculty member' mean Professors, Associate Professors, Assistant Professors, Lecturers having prescribed qualifications, engaged whole time by the University for teaching certificate, diploma, degree or post graduate classes and engaged in research and development of technologies and such other persons as may be declared as teachers by the Regulations;

2) The Department of Information Sciences offers BS and MSc undergraduate and graduate level degree programs. 06 Credit Hours FYP shall mandatory for the degree completion in the Final Semester (8th) under following Rules andRegulations.

3) The student shall register the Final Year Project (FYP) of 06 CHs in the 7th regular semester of the degree program withdetail:

a) Course Code = XXXXXXXX
b) Course Title = Final YearProject
c) Credit Hours = 06 (6+0)
d) Total Marks = 200 (To be awarded by the Project ExternalExaminers)
e)

Breakup of Total Marks (FYP)			
Sr. No.	**Marks Awardees**	**Semester**	**Marks**
1)	Internal Examiner evaluation Committee of FYP	7th	50
2)	Supervisor	7th & 8th	40 (20 in each semester)
3)	PMO	7th & 8th	20 (10 in each semester)
4)	Project External Examiner	8th	90
		Total Marks	**200**

4) A Departmental Final Year Project Committee (DFYPC) shall be constituted and notified by the PMO and Coordinator by the second week of 6th Semester for a batch on yearly basis.

5) Considering seniority, the Chairperson/Head of the Department/ Coordinatorshallforwardthename(s)ofa the Professors, Associate Professors, Assistant Professor and Lecturer (Regular / IPFP / TTS / Long Term Contractual) as PMO to the Director/Principal concerned for notification of the DFYPC for a term of one academic year (Fall and Spring Semesters). Considering seniority, the DFYPC shall be rotated on yearlybasis.

6) The PMO shall be given relaxation of a course(3 credit hours) during his / her tenure. The Chairperson / Head of the Department/Coordinator shall get the approval for the same by the Registraroffice.

7) Considering seniority and excluding the convener/members of the DFYPC, the Chairperson / HOD/Coordinator shall notify the names of the eligible regular / long term contractual / IPFP / TTS Faculty members as Supervisors for FYP with the approval of Director/Principal concerned.

8) Considering number of Supervisors the DFYPC shall define the number of student Groups for conducting FYP on equal sharing. However, this is the discretion of HOD/Coordinator to assigned groups to the faculty members (ifrequired).

9) Considering number of student project Groups, the students of the batch having maximum CGP. After 5th Semester shall be notify, soon after the declaration of 5th Semester results by the Controller of Examinations, as Group Leaders by the DFYPC. For example if there are 10 Groups the 10 top most students of the batch shall be the GroupLeaders.

10) The Group Leaders shall be given the opportunity to select their team members; however, number of team members per Group shall be defined by DFYPC as per number of available Supervisors. Maximum 3 students are allowed in a group.

11) The DFYPC notify the names of the Group Supervisors by the Mid Term Examinations of the 6thSemester.

12) The HOD/Coordinator will assign students to the respective supervisors within the department by taking written consentfrom the project group leader along with level of expertise and title/nature of the project. The approval should be taken by the end of 6th semester from the chairperson/HOD/Coordinator.

13) The Chairperson/HOD/Coordinator shall get notify the Group Supervisor allocation by the Director/Principal concerned by the end of 2nd week of the 7thSemester.

14) Considering limited research period (two semesters) the Project Proposal shall be prepared

under the Supervisor concerned on the UE (BoS) notified format by the Groups.

15) Each Group shall submit the Project Proposal to the respective Chairperson/HOD/Coordinator before Mid Term Examinations of the 7th Semester through supervisor.

16) Each Group shall have to successfully defend the Project Proposal before the DFYPC in the presence of the respective Supervisor by the end of the 12th week of the 7thSemester.

17) The respective Chairperson/HOD/Coordinator shall notify the list of the approved projects along with detail of the Groups/students with a copy to the respective Director/Principal soon after endorsement by the DFYPC.

18) The Groups shall submit final draft of the Project in the UE (BoS) notified format in the 12th week of the 8th semester by the stipulated date notified by theDFYPC.

19) The DFYPC shall submit a panel of three Project External Examiners suggested by supervisor and duly approved by the respective Chairperson in the 10thweek of 8th semester. The project External members shall be within UE campuses or outside UE (HEC recognized DIE of Punjab province) or professionals from reputed organization having minimum 16 Years Education with 5 years experience.

20) The notified Project External Examiners shall be invited by the DFYPC between 14th and 15th week of the 8th Semester but not later than last date of the classheld.

21) The Project External Examiners shall evaluate the projects and after conduct of VivaVoce every Group member shall be awarded marks out of 90 separately on the prescribed format. No internal Examiner / assistant shall be appointed. However; DFYPC shall manage and monitor the entire process. The student wise Awards duly signed by the Project External Examiners shall be submitted to the DFYPC.

22) The DFYPC shall submit students wise Awards with their endorsement to the internal incharge of Examinations within two days of the conduct through the Director/Principalconcerned.

23) The Chairperson / HOD/Coordinator concerned shall ensure the uploading of the Awards in the UMS and Examination data base by himself/supervisor within the stipulate date notified by the Controller of Examinations.

REGISTRAR
UNIVERSITY OF EDUCATION

SOPs for Final Year Project (FYP) Evaluation

Department of Information Sciences has three degree programs (MSc IT, BSIT & BSCS) at undergraduate level for which evaluations are performed for FYP.

1. Students register final year project in 7th semester of 6 credit hours (200 marks).
2. Students are then guided for synopsis/proposal writing and are assigned supervisor and advised to submit proposal of the project.
3. First evaluation for proposal is done at 3rd week of 7th Semester as:

   3.1- Designated Project Management Office (PMO)'s representative suggests evaluation committees to Chairperson/Coordinators.

   3.2- Chairperson approves the committees

   3.3- Evaluation performed and result kept by PMO Office

4. Second Evaluation of 50 marks is done at the end of 7th semester and same steps are followed as described in clause 3.
5. Supervisor's marks (i.e. 20) and PMO's marks (i.e. 10) are then allocated.
6. 7th semester evaluation of 80 marks is ended up here.
7. Student register their project in 8th semester
8. 8th semester evaluation of 90 marks is performed at the end of this semester and same procedure, as given in clause 3 is repeated for evaluation.
9. Supervisor and PMO office marks (20 and 10 marks respectively) for 8th semester are then allocated.
10. 8th semester evaluated of 120 marks is completed.
11. 7th semester and 8th semester marks are summed up.
12. Total marks from 200 marks are then shifted to Final Evaluation Proforma and result is then sent to examination department for each student.

Chairperson

Department of Information Sciences

University of Education, Lahore.

Final Year Project Evaluation Form (7th Sem

Project Title: _____

Roll No._____ Student Name:_____

Project Deliverables	Total Marks	Marks Obtained	Remarks
Proposal Document			
Understanding of existing system	○ Accepted ○ Revised ○ Rejected		
Well defined goal & objective			
Conceptual Application/Architecture (Modules)(Work flow of system)			
Presentation Skills			

Evaluator 1 (Name, Signature &Date)._____

Evaluator 2 (Name, Signature &Date)._____

Software Requirements Specification (Sub-Section Total: 20)				
Functional Requirements (Well defined requirement according to proposal)		4		Internal Examiner evaluation Committee of FYP
Interfaces (UI, External, H/W Interface requirements)		4		
Use case descriptions		4		
Use case diagrams		4		
Non-Functional Attributes (Reliability, Availability, Security, Maintainability, Portability, Performance)		4		
Design Document (Sub-Section Total: 20)				
OO Approach	**Structured Approach**			
Class Diagram	Data Flow Diagram	4		
Sequence Diagram	Architectural Diagram	4		
ERD	ERD	4		
DB Schema	DB Schema	4		
Architectural Diagram	State Transition Diagram	4		
Prototype (Sub-Section Total: 10)				
Mockups, Partially Working System (30% implementation)		10		
Supervisor (Sub-Section Total: 20)				Supervisor
(Meetings, project progress)		20		
Project Management Office (Sub-Section Total: 10)				PMO
(Meeting Deadlines, Attending workshops)		10		
Internal Marks		**80**		

Evaluator 1 (Name, Signature &Date)._____

Evaluator 2 (Name, Signature &Date)._____

It is certified that student project is currently registered in the current semester.

Signature of PMO_____ Dated: -_____

Final Year Project Evaluation Form (8th Semester)

Project Title: _____

Roll No:_____ Student Name:_____

Project Deliverables	Total Marks	Obtained Marks	Remarks
Code (Sub-Section Total: 30)			Project External Examiner (Intra/Inter Campus)
Complete running project in compliance with requirement and design document	15		
Run time code modification and understanding	15		
Testing (Sub-Section Total: 20)			Project External Examiner (Inter/Intra Campus)
Test Plan	5		
Test Case Design and implementation	15		
Overall System and Documentation (Sub-Section Total: 40)			Project External Examiner (Inter/Intra Campus)
Project presentation (Including SRS and Design Document)	25		
Standard Template	5		
Overall skill set	10		
Supervisor(Sub-Section Total: 20)			Supervisor
(Meetings, project progress)	20		
Project Management Office(Sub-Section Total: 10)			PMO
(Meeting Deadlines, Attending workshops)	10		
Total Marks	**120**		

Evaluator 1 (Name, Signature &Date)._____

Evaluator 2 (Name, Signature &Date)._____

It is certify that student project is register in current semester. Moreover, the student/group has passed previous semester project evaluation.

Signature of PMO_____ Dated: -_____

Project Proposal Summary

Registration Number	Student Name
123456	XYZ
123456	XYZ
123456	XYZ

Project Title: **Manage the Constituency: Android App**

Target Audience: Politicians

Project Description:

Politics in Pakistan is very complex. A candidate who needs to contest in an election has to do a lot on several fronts. The tribe system, party, sect and many more play a major role to get a victory. Furthermore, the union council level demography on the maps is very much need of any contestant. Therefore, many Politicians find difficulty to remember previous data or records of their previous elections in a particular constituency, supporters, political strengths and weaknesses. This application is specifically designed to help Politicians to manage their records, achievements, granting, won elections and their statistics about it on the Google map up to union council and village level. This app provides complete confidentiality to its users. User will register himself, and thus upon successful login he will be able to select his concerned area.

Tools& Technologies

- Android Studio
- Google Map (API)
- Photoshop
- Microsoft Visio
- SQL server

----------TO BE FILLED BY THE COMMITTEE----------

- Accepted
- Revised
- Rejected

Remarks (If any):

Name & Signatures: _____

Name & Signatures: _____

Name & Signatures: _____